Final Project Report

Computer Networks (CS-307)

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FINAL PROJECT REPORT

Design and Configuration of the Complete Network for The University of Faisalabad (TUF)

1. Introduction

Computer networks are essential in educational institutions for communication, research, administration, and sharing academic resources. The University of Faisalabad requires a reliable and scalable network infrastructure connecting Amin Campus and Saleem Campus.

This project presents the design and implementation of a complete university network using Cisco Packet Tracer. The project includes VLAN segmentation, inter-VLAN routing, DHCP services, WAN connectivity and cloud email server communication.

The network follows a hierarchical architecture consisting of core, distribution, and access layers.

2. Project Topology Overview:

The project topology consists of:

- Routers in Core Layer
- Layer 3 Switches in Distribution Layer
- Cisco 2960 Switches in Access Layer
- PCs, printers, and servers connected to access switches
- Separate cloud router and email server Both campuses are connected

through serial WAN links.

3. Network Layers:

- **Core Layer**

The core layer contains routers responsible for communication between campuses and external cloud services.

- **Distribution Layer**

Layer 3 switches are used for VLAN routing and network management.

- **Access Layer**

Cisco 2960 switches connect end devices such as:

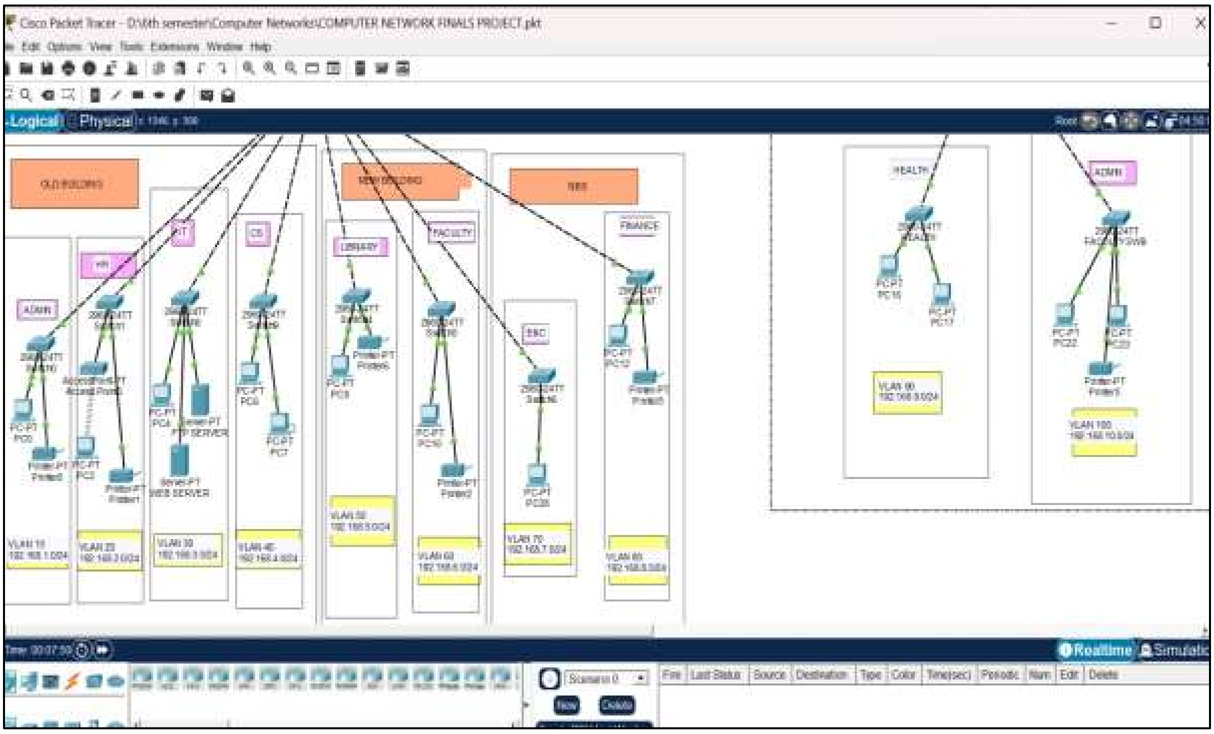
- PCs
- Printers
- Servers

4. Physical Connections:

Connection	Cable Type
Router ↔ Router	Serial DCE
Router ↔ Layer 3 Switch	Copper Straight Through
Layer 3 Switch ↔ Access Switch	Copper Cross Over
PC ↔ Switch	Copper Straight Through

5. VLAN Design:

A total of 10 VLANs were implemented.



Saleem Campus VLANs:

VLAN ID	Department	Network
90	HEALTH	192.168.9.0/24
100	ADMIN	192.168.10.0/24

Amin Campus VLANs:

VLAN ID	Department	Network
10	Admin	192.168.1.0/24
20	HR	192.168.2.0/24
30	IT Department	192.168.3.0/24
40	CS	192.168.4.0/24
50	Library	192.168.5.0/24
60	Faculty	192.168.6.0/24
70	E&C	192.168.7.0/24
80	FINANCE	192.168.8.0/24

6. WAN Addressing Scheme:

Connection	Network
Amin Campus ↔ Saleem Campus	10.10.10.0/30
Amin Campus ↔ Cloud Router	10.10.10.4/30
Cloud Router ↔ Email Server	20.10.10.0/24

7. Amin Campus Router Configuration:

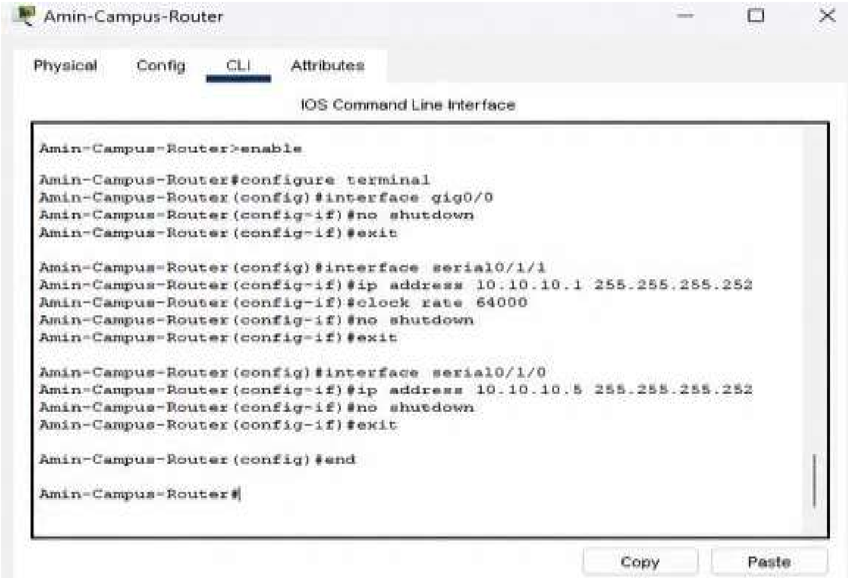
Basic Configuration:

enable configure
terminal

interface gig0/0 no
shutdown

interface serial0/1/1 ip address
10.10.10.1 255.255.255.252 clock rate
64000 no shutdown

interface serial0/1/0 ip address 10.10.10.5 255.255.255.252 no shutdown



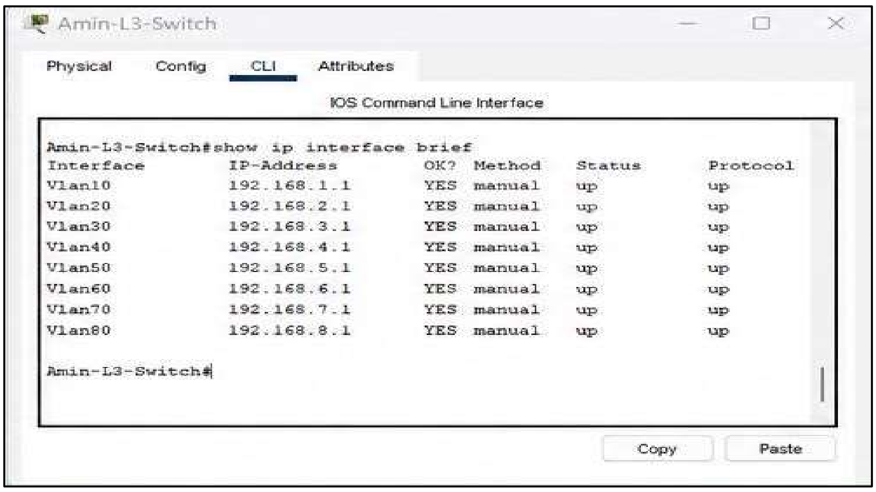
Inter-VLAN Routing Configuration

interface gig0/0.10 encapsulation dot1Q 10 ip address 192.168.1.1 255.255.255.0

interface gig0/0.20 encapsulation dot1Q 20 ip address 192.168.2.1 255.255.255.0

interface gig0/0.30 encapsulation dot1Q 30 ip address 192.168.3.1 255.255.255.0

The same configuration was repeated for all VLANs.



8. DHCP Configuration:

DHCP Pool for VLAN 10

service dhcp

ip dhcp pool ADMIN network
192.168.1.0 255.255.255.0 default-
router 192.168.1.1 dns-server
192.168.1.1

DHCP pools were configured for all VLANs.

9. Cloud Router Configuration:

enable configure
terminal

interface serial0/1/0 ip address
10.10.10.6 255.255.255.252 clock rate
64000 no shutdown

interface gig0/0 ip address
20.10.10.1 255.255.255.0 no
shutdown

10. Saleem Campus Router Configuration:

enable configure
terminal

interface serial0/1/0 ip address
10.10.10.2 255.255.255.252 no
shutdown **VLAN 90 Configuration**

interface gig0/0.90 encapsulation
dot1Q 90 ip address 192.168.9.1
255.255.255.0 **VLAN 100
Configuration**

interface gig0/0.100 encapsulation
dot1Q 100 ip address 192.168.10.1
255.255.255.0

11. Layer 3 Switch Configuration:

Trunk Port Configuration

interface gig1/0/1 switchport trunk

encapsulation dot1Q switchport

mode trunk **Access Port Configuration**

interface gig1/0/2 switchport
mode access switchport
access vlan 10

The same steps were repeated for all VLANs.

12. Access Layer Switch Configuration:

Admin Switch

enable configure
terminal

interface range fa0/1-24

switchport mode access

switchport access vlan 10 **HR Switch**

interface range fa0/1-24
switchport mode access switchport
access vlan 20

The same configuration was repeated for all remaining switches.

13. Email Server Configuration

A static IP address was assigned to the email server.

Setting	Value
IP Address	20.10.10.2
Subnet Mask	255.255.255.0
Default Gateway	20.10.10.1

Ping testing between the cloud router and email server was successful.

14. Verification Commands

The following commands were used for verification:

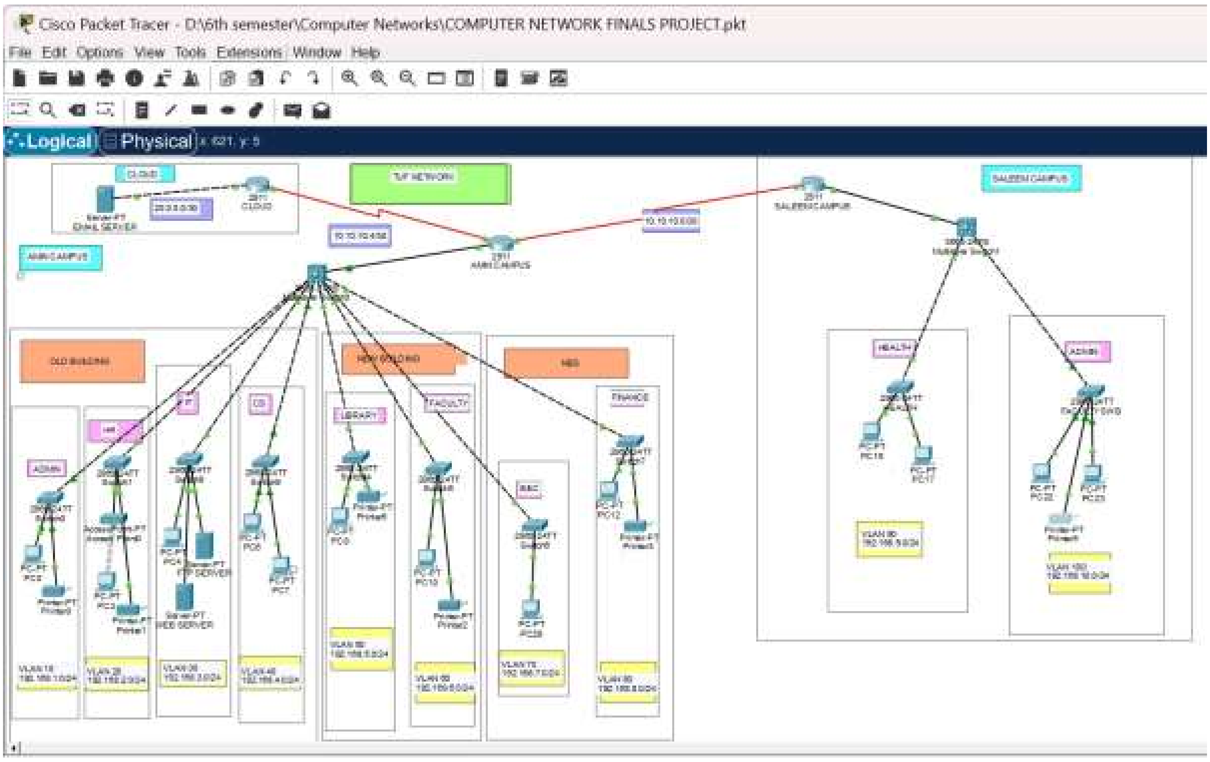
show ip interface brief
show ip route show ip
protocols

15. Testing and Verification

The following tests were performed successfully.

Test	Result
Ping within same VLAN	Successful
Ping between different VLANs	Successful
Communication between campuses	Successful
DHCP IP assignment	Successful
Email server connectivity	Successful

16. Final network:



17. Conclusion

This project successfully designed and implemented a complete network infrastructure for The University of Faisalabad using Cisco Packet Tracer.

The network connected Amin Campus and Saleem Campus through WAN serial links and dynamic routing protocols. VLAN segmentation improved security and reduced unnecessary traffic.

The final implementation provided a scalable, manageable, and reliable campus network suitable for university operations.